

Customer Profiling and Value Segmentation Analysis.

Executive Summary

This analysis was conducted to develop a practical customer profiling framework that segments the bank's customers into meaningful business tiers and identifies opportunities to improve customer value and retention.

The dataset contained 1,006 customer records covering demographics, financial behaviour, product ownership, transaction activity and digital engagement. Before profiling customers, the dataset was assessed for quality issues including duplicate records, unrealistic values and inconsistent categorical fields. After cleaning, the data was suitable for analysis and customer segmentation.

Rather than creating statistical clusters that are difficult for business users to interpret, a transparent weighted scoring model was developed using five key dimensions of customer value:

- Financial strength
- Transaction behaviour
- Product ownership
- Digital engagement
- Relationship history

Each customer received a score and was classified into one of five business tiers:

- Platinum
- Gold
- Silver
- Bronze
- Black

The analysis shows clear behavioural differences between customer groups. Platinum customers maintain significantly higher balances, own more banking products and engage with the bank more frequently across both physical and digital channels. In contrast, Black tier customers display low balances, limited product usage and weak engagement, indicating significant growth opportunities.

The analysis also identifies realistic actions that could move customers into higher-value segments through targeted campaigns, improved digital engagement and cross-selling initiatives. For Platinum customers, early warning indicators were developed to detect declining engagement before valuable relationships are lost.

Finally, recommendations are provided for incorporating external demographic, credit bureau and macroeconomic data to strengthen future customer profiling while remaining compliant with Sri Lanka's Personal Data Protection Act (PDPA).

1. Project Objective

The objective of this analysis was to develop a customer segmentation framework that enables the bank to:

- understand the value of each customer,
- identify opportunities to increase customer value,
- improve customer retention,
- support targeted marketing campaigns, and
- provide a practical framework that business teams can easily understand and maintain.

Rather than relying solely on demographic information, the segmentation combines financial behaviour, product usage and customer engagement to provide a more complete view of customer value.

2. Technical Stack

NumPy	Numerical calculations
Scikit-learn	Min-Max normalization and feature scaling
Matplotlib	Data visualization
Power BI	Interactive dashboard development
Microsoft Excel	Data validation and output reporting
Google Colab	Python development environment

3. Data Preparation

Before beginning the analysis, the dataset was assessed to identify issues that could affect the quality of the results.

The following data quality checks were performed:

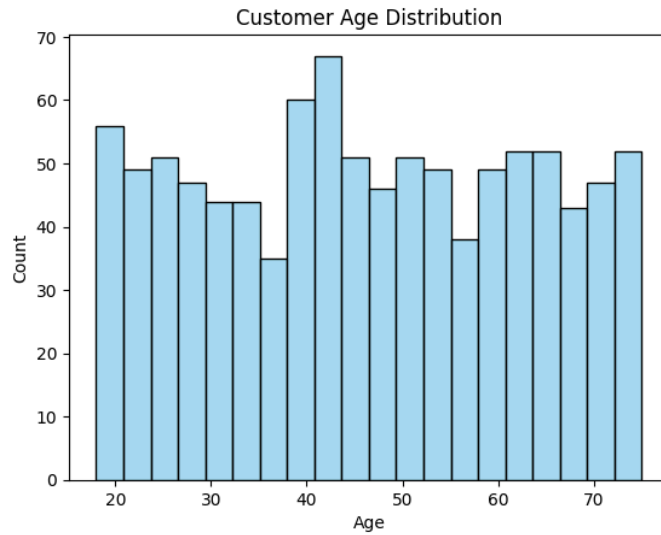
Cleaning Activity	Purpose
Duplicate removal	Prevent double counting
Invalid age removal	Improve demographic accuracy
Negative value removal	Correct financial data quality issues
Missing value treatment	Maintain complete analysis dataset
Category standardization	Ensure consistency

4. Exploratory Data Analysis (EDA)

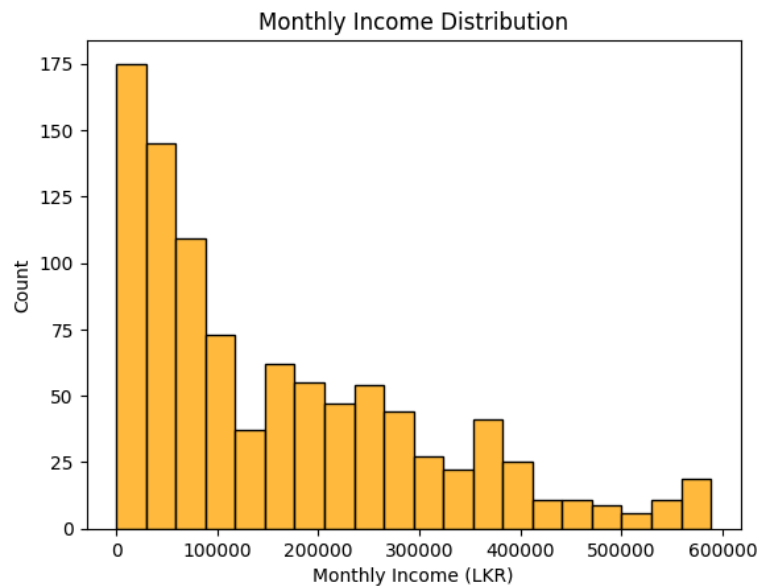
Exploratory analysis was performed to understand the customer base before developing the segmentation model.

The analysis focused on:

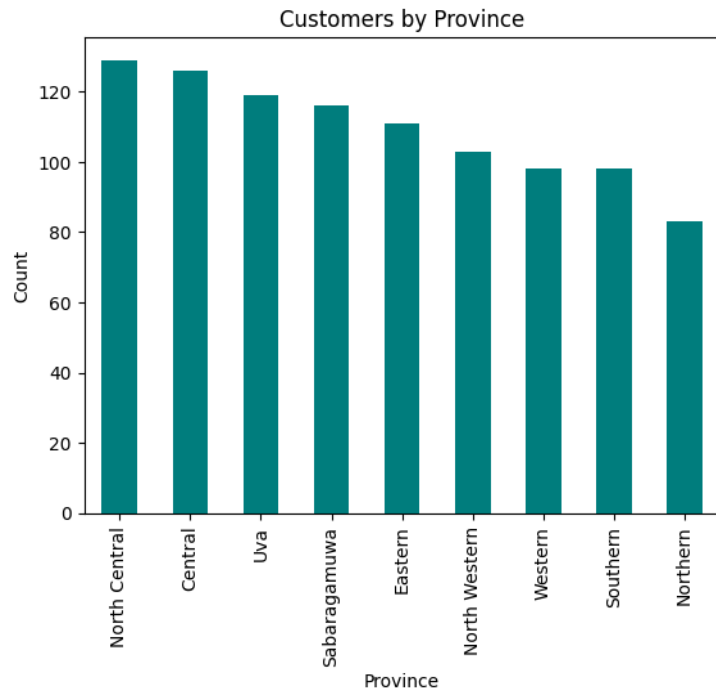
- customer age distribution
- income distribution
- geographic distribution
- product ownership
- transaction behaviour
- digital engagement



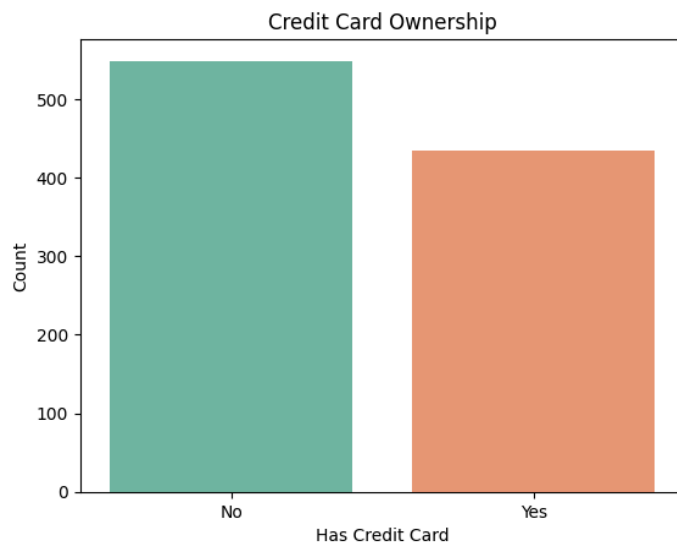
Most customers fall within the economically active age range, suggesting a customer base with strong long-term banking potential.



Income is positively skewed, with most customers earning below LKR 200,000 per month while a smaller group represents significantly higher-income individuals.



Customer distribution varies across provinces, indicating potential regional differences in income levels and banking behaviour.



Approximately half of the customers currently own a credit card, suggesting opportunities for additional product penetration.

Overall, the dataset contains sufficient variation across financial and behavioural attributes to support meaningful customer segmentation.

5. Defining Active and Inactive Customers

Before assigning customer tiers, it was important to distinguish customers who actively engage with the bank from those who do not.

A customer was classified as **Active** when all of the following conditions were satisfied:

- Not marked as dormant
- Last transaction occurred within the previous 90 days
- At least 10 transactions per month
- At least 3 mobile application logins per month

Customers who did not satisfy these criteria were classified as **Inactive**.

This definition combines financial activity with digital engagement, providing a broader measure of customer participation than transaction volume alone.

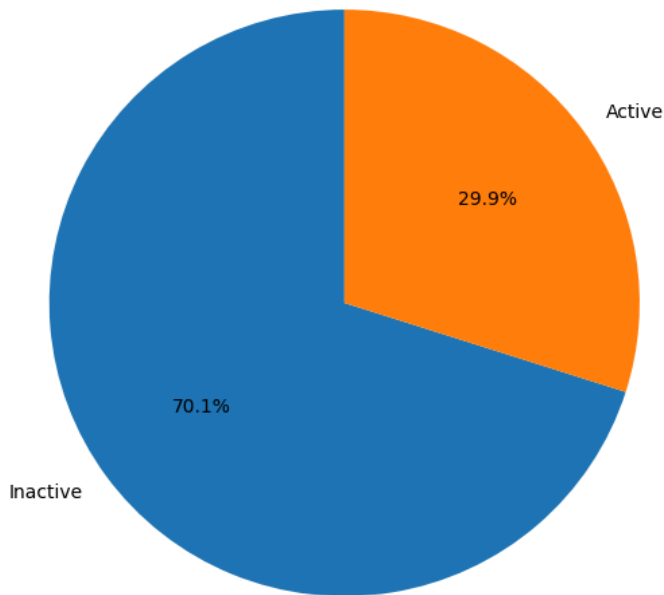
Approximately **30%** of customers met all activity criteria.

```
#..... Defining Active vs Inactive Customers.....

df['Active'] = np.where(
    (df['Dormancy Flag']=="No") &
    (df['Last Transaction Date (Days Since)'] <= 90) &
    (df['Total Transactions per Month'] >= 10) &
    (df['Mobile App Logins per Month'] >= 3),
    "Active", "Inactive"
)
print(df['Active'].value_counts())

... Active
Inactive    689
Active      294
Name: count, dtype: int64
```

Active vs Inactive Customer Distribution



6. Feature Engineering & Scoring Model

We grouped features into five dimensions: Financial, Transaction, Product, Digital, Relationship. For each, we selected representative metrics:

Financial Score: Average Balance (3-month), Income, Total Savings Balance.

Transaction Score: Total Transactions per Month, Transaction Days per Month.

Product Score: Total Products Held (savings, card, loan, FD, insurance, investment).

Digital Score: Mobile App Logins per Month (and digital fund transfers).

Relationship Score: Customer Tenure (months with bank) [optionally Complaints or Churn flag].

All numeric features were MinMax-scaled. We then combined them with hand-picked weights reflecting business importance (roughly): Financial 35%, Transaction 25%, Product 20%, Digital 10%, Relationship 10%.

Summary table:

Dimension	Weight	Business Rationale	Variables
Financial Strength	35%	Primary indicator of customer value	Income, Savings Balance, Average Balance
Transaction Behaviour	25%	Measures engagement	Monthly Transactions, Transaction Days
Product Ownership	20%	Indicates relationship depth	Number of Products Held
Digital Engagement	10%	Reflects modern banking behaviour	Mobile App Usage
Customer Relationship	10%	Rewards long-term loyalty	Customer Tenure

Continuous variables were normalised before calculating the weighted customer score.

To ensure that variables with different measurement scales could be combined fairly, all numerical features were normalised using the Min-Max Scaling technique implemented through Python's `MinMaxScaler()` function. The values were transformed into a common range between 0 and 1.

This prevents high-value variables such as income and account balances from dominating the scoring model and allows all customer attributes to contribute proportionally to the final customer score.

Financial Score: Calculated using the average of normalized income, average balance, and savings balance. This represents the customer's financial strength.

Transaction Score: Calculated using normalized transaction volume and transaction frequency, reflecting customer activity and engagement.

Product Score: Based on the normalized number of products held by each customer, indicating relationship depth and cross-selling potential.

Digital Score: Derived from mobile app login activity, representing digital engagement.

Relationship Score: Based on customer tenure with the bank, representing long-term relationship value

A weighted customer score was calculated to measure overall customer value. Each customer received a score between **0 and 100** by combining five behavioural and financial dimensions.

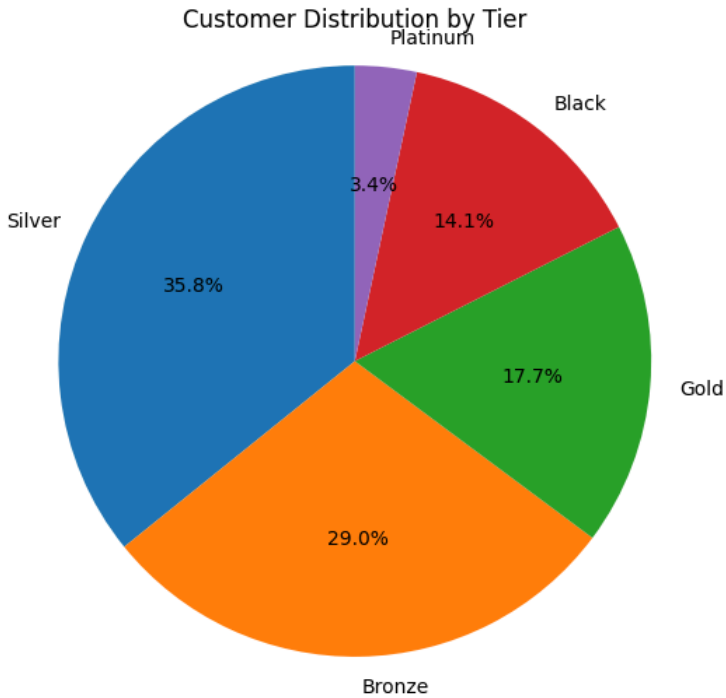
Customer Score Formula:

Customer Score= (35%×Financial Score) + (25%×Transaction Score) +(20%×Product Score) + (10%×Digital Score) + (10%×Relationship Score)

7. Customer Tier Classification

Based on the overall customer score, customers were assigned to one of five business tiers.

Tier	Customer Score
Platinum	≥60
Gold	50–59
Silver	40–49
Bronze	30–39
Black	<30



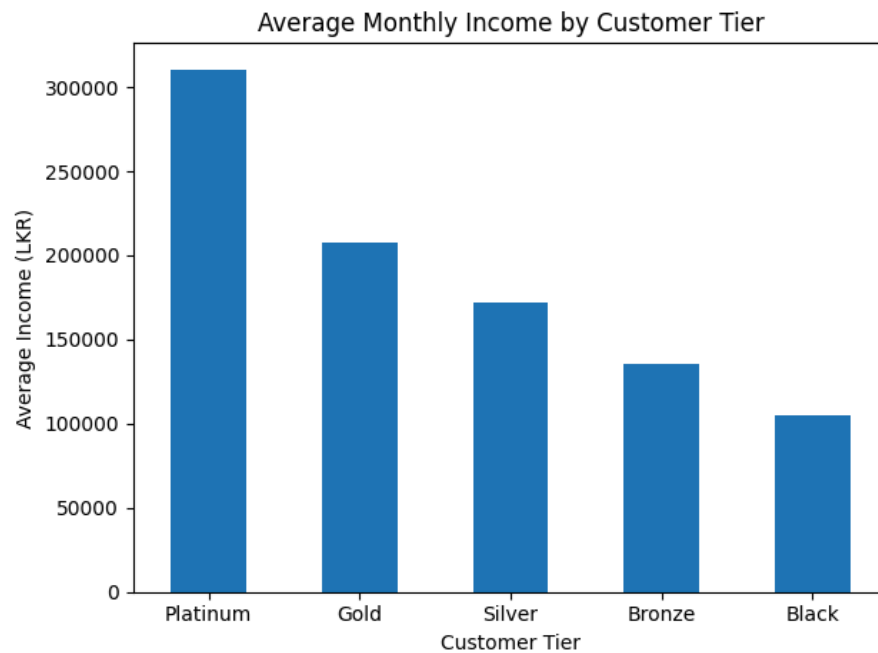
The distribution demonstrates a balanced customer portfolio, with the majority of customers falling within the Silver and Bronze segments.

8. Customer Profiles

Tier	Key Characteristics
Platinum	Highest income, largest balances, multiple products, strong digital engagement, highly active
Gold	Financially strong customers with good engagement and potential to become Platinum
Silver	Moderate balances and engagement with clear cross-selling opportunities
Bronze	Limited balances and product ownership but potential for growth
Black	Low engagement, low balances and limited banking relationship

...	Tier	Avg Income (LKR)	Avg Balance (LKR)	Avg Products	Total Txn/mo	\
	Platinum	310557.0	410321.0	4.0	54.0	
	Gold	208010.0	383992.0	3.0	52.0	
	Silver	172307.0	286031.0	3.0	47.0	
	Bronze	135491.0	187324.0	2.0	43.0	
	Black	104657.0	129718.0	2.0	29.0	

	Tier	App Logins/mo
	Platinum	42.0
	Gold	36.0
	Silver	32.0
	Bronze	27.0
	Black	20.0



These profiles provide clear guidance for relationship management and targeted marketing.

9. Tier Uplift Strategy

Rather than applying identical marketing campaigns to every customer, each tier should receive interventions that address the largest barrier preventing progression.

Current Tier	Largest Barrier	Recommended Actions
Black	Low balances and limited products	Salary credit campaigns, savings incentives, entry-level products
Bronze	Low balances and weak digital engagement	Digital banking education, bundled products
Silver	Moderate financial activity	Investment products, insurance, credit expansion
Gold	Limited wealth products	Dedicated relationship management, premium investment solutions

These actions focus on improving customer lifetime value while increasing product penetration.

10. Platinum Customer Retention

Platinum customers generate the greatest long-term value and therefore require proactive monitoring.

Early warning indicators include:

- declining account balances
- reduced transaction frequency
- fewer mobile application logins
- increasing dormancy
- complaints or service issues
- churn risk indicators

A Platinum customer is considered **stable** when financial activity and engagement remain consistent over time.

A customer is considered **in decline** when multiple engagement indicators deteriorate over consecutive months.

Recommended retention initiatives include:

1. Dedicated relationship managers
2. Personalised financial reviews
3. Exclusive investment products

4. Loyalty rewards
5. Early intervention based on predictive alerts

This proactive approach enables the bank to retain high-value customers before disengagement becomes permanent.

```
Platinum Retention & Risk Signals

plat_df = df_scaled[df_scaled['Tier']=="Platinum"]
at_risk = plat_df[(plat_df['Last Transaction Date (Days Since)'] > 60) |
                  (plat_df['Total Transactions per Month'] < 15) |
                  (plat_df['Mobile App Logins per Month'] < 5)]
print("Potential Platinum decline signals:", at_risk.shape[0])

Potential Platinum decline signals: 33
```

11. External Data Enrichment

Internal banking data provides only a partial view of customer behaviour.

Future customer profiling could be strengthened by integrating carefully governed external data sources.

External Data	Business Value
Household Income & Census Data	Regional income benchmarking
Credit Bureau (CRIB)	External debt and credit risk assessment
Digital Adoption Statistics	Better interpretation of digital engagement
Macroeconomic Indicators	Context for balance and spending changes
Consumer Behaviour Data	Improved understanding of spending patterns

Any external data integration should comply with Sri Lanka's Personal Data Protection Act (PDPA) and ensure that customer privacy, consent and ethical data usage remain central to all analytical processes.

12. Key Business Recommendations

Based on the analysis, the following actions are recommended:

- Adopt the proposed customer scoring model as a standard segmentation framework.
- Prioritise customer growth initiatives for Bronze and Silver segments, where the largest opportunities exist.
- Introduce automated monitoring for Platinum customer engagement to identify early signs of attrition.
- Increase digital banking adoption through personalised campaigns.
- Expand customer profiling using external demographic and credit bureau data while maintaining regulatory compliance.

Implementing these recommendations would enable the bank to improve customer retention, increase product penetration and allocate marketing resources more effectively.

13. Conclusion

This analysis demonstrates how customer data can be transformed into practical business intelligence through structured profiling and transparent scoring.

Rather than creating segmentation based purely on statistical groupings, the proposed framework aligns customer behaviour with measurable business value. The resulting customer tiers provide a clear foundation for targeted marketing, relationship management and retention strategies, while remaining easy for both technical and non-technical stakeholders to understand.

The framework is also scalable. As additional customer behaviour and external data become available, the scoring model can be refined to improve prediction accuracy and support more personalised banking strategies.

11. Appendix :

Python code (google colab)

<https://colab.research.google.com/drive/1AiYarJdN1Qd1GjPqaG3diBAAuJCwaNLE?usp=sharing>

```
#.....Understand the data set.....

import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

pd.set_option("display.max_columns", None)

df = pd.read_excel("Customers_Data.xlsx")

print(df.shape)

df.head()

df.info()

df.describe(include="all")
```

... (1006, 63)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1006 entries, 0 to 1005
Data columns (total 63 columns):

#	Column	Non-Null Count	Dtype
0	Customer ID	1006 non-null	object
1	Full Name	1006 non-null	object
2	Age	1006 non-null	int64
3	Gender	1006 non-null	object
4	Province	1006 non-null	object
5	City	1006 non-null	object
6	Marital Status	1006 non-null	object
7	Employment Status	1006 non-null	object
8	Occupation Category	1006 non-null	object

```
Feature Engineering & Scoring Model.....

from sklearn.preprocessing import MinMaxScaler

features = ['Estimated Monthly Income (LKR)', 'Avg Balance Last 3 Months (LKR)', 'Savings Account Balance (LKR)',
            'Total Transactions per Month', 'Transaction Days per Month',
            'Total Products Held',
            'Mobile App Logins per Month',
            'Customer Tenure (Months)']

scaler = MinMaxScaler()
df_scaled = df.copy()
df_scaled[features] = scaler.fit_transform(df[features])

# component scores (0-100 scale)
'Financial Score' = df_scaled[['Estimated Monthly Income (LKR)', 'Avg Balance Last 3 Months (LKR)', 'Savings Account Balance (LKR)']].mean(axis=1)*100
'Transaction Score' = df_scaled[['Total Transactions per Month', 'Transaction Days per Month']].mean(axis=1)*100
'Product Score' = df_scaled[['Total Products Held']]*100
'Digital Score' = df_scaled[['Mobile App Logins per Month']]*100
'Relationship Score' = df_scaled[['Customer Tenure (Months)']]*100

# total score (0-100)
'Customer Score' = (
    df_scaled['Financial Score'] +
    df_scaled['Transaction Score'] +
    df_scaled['Product Score'] +
    df_scaled['Digital Score'] +
    df_scaled['Relationship Score']
)

df_scaled[['Financial Score', 'Customer Score']].describe().round(1)
```

... Financial Score Customer Score

	Financial Score	Customer Score
count	983.0	983.0
mean	39.9	41.6
std	19.4	10.3

[]



```
#...7. Customer Tier Classification...
```

```
def assign_tier(score):  
    if score >= 60: return "Platinum"  
    elif score >= 50: return "Gold"  
    elif score >= 40: return "Silver"  
    elif score >= 30: return "Bronze"  
    else: return "Black"  
  
df_scaled['Tier'] = df_scaled['Customer Score'].apply(assign_tier)  
print(df_scaled['Tier'].value_counts())
```

```
import matplotlib.pyplot as plt  
  
tier_counts = df_scaled['Tier'].value_counts()  
  
plt.figure(figsize=(6,6))  
plt.pie(  
    tier_counts,  
    labels=tier_counts.index,  
    autopct='%1.1f%%',  
    startangle=90  
)  
  
plt.title("Customer Distribution by Tier")  
plt.axis('equal')  
plt.show()  
  
# import os  
  
# # Folder where you want to save the file  
# folder path = "Outputs"
```